Google Cloud AI Builder Cup 2026 · **Future of Work**

DGUI HUB NET

An agent fabric where individuals route real jobs to each other — executed by **Gemini** on **Cloud Run**, and sealed with evidence you can check.



Gemini | Cloud Run | Firestore | Governed execution

We hand real work to software we cannot verify.

- Agents run quietly in silos — no portable identity, no proof of who ran what.
- Tool access is all-or-nothing — a leaked API key is full access.
- Output is trusted on faith — there is no gate that can fail.

LOCAL TRUTH

Two professionals cannot safely route a job to each other's agents today without handing over credentials.

GLOBAL COST

As agentic work scales, unverifiable execution becomes uninsurable — trust, not compute, is the bottleneck.

THE SOLUTION

A governed agent fabric — portable, gated, provable.

01 · TRUST BOUNDARY

Jobs start from a transport policy, not a key: private person-to-person work is refused over the public internet, allowed on the LAN.

02 · RIGHT AGENT

A persona-match gate attaches the correct agent blueprint before execution.

03 · GEMINI EXECUTION

The job runs on Gemini; the dock agent it mints is what closes the evidence gate.

04 · PROOF

Three layers of validation — structural, governance, operational — must all pass. Mintable only if they do.

LIVE DEMO

One flow: reject → route → approve → execute → prove.

dgui-hubnet-google · https://{{DEPLOYED_URL}}

```
# 1 · public dock → refused
connect(i2i, https://203.0.113.42:3198)  →  REJECTED: phase-1 i2i is LAN-only
# 2 · LAN dock → accepted
connect(i2i, http://127.0.0.1:3198)      →  ok:true  state: requested
approve(Bella) → connected   assign(Alex) → REQUESTED   advance → MATCHED
# 3 · evidence gate BEFORE execution
validate  →  L1 PASS · L2 PASS · L3 FAIL   (no dock agent yet)
# 4 · Gemini execution mints the dock agent
execute   →  engine: gemini · model: gemini-2.5-flash · tokens: N · latency: N ms
# 5 · gate closes
validate  →  L1 PASS · L2 PASS · L3 PASS   →  mintable
```

ARCHITECTURE · GOOGLE CLOUD

Thin edge, Google-native core.

Individual A
Docking Station
HUB NET client

Cloud Run
HUB NET API
transport guard
evidence gate

Gemini
@google/genai
dock-agent minting
job execution

Firestore
connections
assignments
evidence log

- Stateless service; state lives in Firestore (memory fallback for offline demo).
- Gemini key held in **Secret Manager** — never baked into the image.
- Deploy: gcloud builds submit --config services/hubnet-google/cloudbuild.yaml

GENAI DEPTH

Gemini isn't decoration — it's the execution layer.

DOCK AGENT

Minted by Gemini per assignment; its identity is the operational fact L3 requires.

GOVERNED PROMPT

Role, company, user and assignment bound into the prompt — an auditable, reproducible unit of work.

METERED

Tokens and latency captured per job → a real, billable KPI surface.

GRACEFUL DEGRADATION

Offline it falls back to a deterministic stub so the demo and CI never hard-fail.

DIFFERENTIATOR

The gate can fail — and we show it.

- Most agent demos only show success. Our evidence gate fails by design until the work is real.
- A physical device key (DGUI Tufty) anchors identity at the edge — work is gated, not just authorised.
- Transport policy is enforced, not advisory — the guard rejects, it doesn't warn.

IMPACT & SCALABILITY

From two people to a fabric of millions.

1	→	**LAN team**	→	**Federated fabric**
INDIVIDUAL ROUTES A JOB		trusted local execution		cross-company routing with the same gate

IMPACT

Skilled people monetise agent work safely; organisations get provable automation instead of blind trust.

SCALABILITY

Cloud Run scales to zero and out; Firestore and the evidence gate are stateless-fronted, so the fabric grows without a rewrite.

BUSINESS CASE & ROADMAP

Monetise trust, not just tokens.

NOW

Phase-1 i2i on LAN · Gemini execution · three-layer evidence gate. Deployed on Cloud Run.

NEXT

WebSocket/WebRTC adapter → remote i2i; team & company docks; KPI billing on metered tokens.

REVENUE

Per-mintable-job fee · evidence-as-a-service for compliance · enterprise federated docks.

MOAT

The gate + physical device key: competitors ship agents; we ship accountable agents.

DGUI HUB NET

The future of work is portable — and provable.

TEAM

{{TEAM}}

LINKS

Live: {{DEPLOYED_URL}}
Repo: {{REPO_URL}}
Video: {{VIDEO_URL}}